Management Science Assignment 1

Total 9 problems

Department: _____

Student ID: _____

Name: _____

Problem 1

Company C produces Products 1 and 2. Each unit of Product 1 earns a profit of KRW 40,000, and each unit of Product 2 earns a profit of KRW 30,000. The company wants to maximize its total daily profit.

Each unit of Product 1 requires 2 machine-hours and 1 labor-hour. Each unit of Product 2 requires 1 machine-hour and 1 labor-hour. The company has 100 machine-hours and 80 labor-hours available per day. All units produced can be sold.

Formulate a linear programming model to maximize total daily profit.

Problem 2

A workshop produces Products X and Y. Quantities x and y are continuous and nonnegative. Its weekly profit, in units of KRW 10,000, is described by the following model:

$$\text{Maximize} \quad 3x + 2y$$

subject to

$$x + y \le 70 \quad \text{(labor hours)}$$

$$2x + y \le 100 \quad \text{(machine hours)}$$

$$x, y \ge 0$$

Solve this model graphically. Show the feasible region and identify the unique optimal solution and the optimal objective value on your graph.

Problem 3

F Electronics produces standard amplifiers (s) and high-performance amplifiers (h). Each standard amplifier earns KRW 30,000 in profit, and each high-performance amplifier earns KRW 50,000. Quantities are continuous and nonnegative. The daily profit-maximization model is:

Maximize $3s + 5h$ (unit: KRW 10,000)

subject to

$s + 2h \leq 180$ (assembly hours)

$s + h \leq 140$ (testing hours)

$h \leq 45$ (semiconductor chips)

$s, h \geq 0$

Consider the production plan $s = 100$ and $h = 40$, with continuous quantities**. (1) Calculate the unused amount of each resource (i.e. slack variables), (2) use it to classify each resource constraint as binding or nonbinding, and (3) discuss the managerial implications of the slack variables and binding or nonbinding constraints.**

Problem 4

Write complete Python code using CVXPY to solve the following model from Problem 3.
Here, s and h are the daily production quantities of standard and high-performance amplifiers. Print the optimization status, optimal objective value, and optimal values of s and h.

Maximize $z = 3s + 5h$ (unit: KRW 10,000)

subject to

$s + 2h \le 180$ (assembly hours)

$s + h \le 140$ (testing hours)

$h \le 45$ (semiconductor chips)

$s, h \ge 0$

Problem 5

Consider the following two models. In both models, x and y are continuous and nonnegative.

Model A

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & x + y \le 5, \\ & x \ge 4, \\ & y \ge 3 \end{aligned}$$

Model B

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & x - y \le 4, \\ & x \ge 0, \\ & y \ge 0 \end{aligned}$$

A student claims, “Neither model has a finite optimal solution, so both models must be infeasible.” **Evaluate this claim by giving the correct status of each model and explaining the mathematical distinction.**

Problem 6

Use the model from Problem 2 shown below. Suppose the coefficient of x changes from 3 to c, while the coefficient of y remains 2 and all constraints remain unchanged. The original optimal solution is (x, y) = (30, 40).

Maximize $3x + 2y$

subject to

$x + y \leq 70$ (labor hours)

$2x + y \leq 100$ (machine hours)

$x, y \geq 0$

Find the range of c for which this point remains optimal.

Information for Problems 7 to 9

Company Y produces cement blocks A, B, C, and D. Production quantities are continuous and measured in tons. Products pass through mixing, molding, testing, and drying; drying capacity is unlimited. Monthly capacities are 800 mixing hours, 700 molding hours, and 500 testing hours. Profit per ton is KRW 40,000, 80,000, 50,000, and 100,000, respectively. Company policy requires the quantity of A to be at least the quantity of B.

The following model maximizes total monthly profit subject to the resource and policy constraints.

$$\text{Maximize} \quad 4A + 8B + 5C + 10D \quad \text{(unit: KRW 10,000)}$$

subject to

$$2A + B + C + 3D \le 800 \quad \text{(mixing hours)}$$

$$A + 2B + C + 3D \le 700 \quad \text{(molding hours)}$$

$$A + B + C + 2D \le 500 \quad \text{(testing hours)}$$

$$A - B \ge 0 \quad \text{(policy constraint)}$$

$$A, B, C, D \ge 0$$

The table below provides optimized and reoptimized results for Problems 7 to 9. Each row is a separate scenario; all other original model data remain unchanged.

Scenario	A	B	C	D	Maximum profit (KRW 10,000)
Original model	200	200	100	0	2,900
Mixing capacity increased to 801 hours	200	200	100	0	2,900
Molding capacity increased to 701 hours	201	201	98	0	2,902
Testing capacity increased to 501 hours	199	199	103	0	2,903
Production of D fixed at 1 ton	199	199	100	1	2,898

Problem 7

(1) Calculate the shadow price of three resources (mixing, molding, testing) in KRW per hour.
(2) Explain what each shadow price means for Company Y. Compare the value of the resources based on the shadow price.

Problem 8

(1) Calculate Product D’s reduced cost in KRW per ton. (2) Explain why Company Y does not produce D in the original optimal plan and under which circumstance the company **may produce** D.

Problem 9

Company K offers Company Y up to 120 additional molding hours at KRW 30,000 per hour; fractional hours may be purchased. The original unit profits already deduct an in-house variable molding cost of KRW 20,000 per hour, which is avoided for each outsourced hour. Outsourcing adds usable molding capacity and changes no other original model data.

Molding capacity is originally 700 hours. Its current shadow price remains valid when total molding capacity is between 500 and 750 hours, with all other data fixed.

(1) Calculate **the net additional cost** and **marginal net benefit** when Company Y outsource one hour of molding to Company K. (2) **Recommend the largest number of hours** to outsource and **calculate the resulting total profit gain**.